

Example: which order of differentiation will calculate f_{xy} faster; x first or y first ①

$$f(x,y) = x^2 + 5xy + \sin x + 7e^x$$

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} (x^2 + 5xy + \sin x + 7e^x) \\ &= 5x\end{aligned}$$

$$\frac{\partial f}{\partial x \partial y} = \frac{\partial}{\partial x} (5x) = 5$$

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} (x^2 + 5xy + \sin x + 7e^x) \\ &= 2x + 5y + \cos x + 7e^x\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} (2x + 5y + \cos x + 7e^x) \\ &= 5\end{aligned}$$

Example, Find the value of $\frac{\partial z}{\partial x}$ at the point $(1, 1, 1)$ if the equation $xy + z^3x - 2yz = 0$

$$\frac{\partial}{\partial x} (xy + z^3x - 2yz) = \frac{\partial}{\partial x} (0) = 0$$

~~$$y + z^3 - 2y \frac{\partial z}{\partial x} = 0$$~~

$$\frac{\partial}{\partial x} (xy) + \frac{\partial}{\partial x} (z^3x) - 2y \frac{\partial z}{\partial x} = 0$$

$$y + \left[z^3 + 3z^2 \frac{\partial z}{\partial x} x \right] - 2y \frac{\partial z}{\partial x} = 0$$

$$y + z^3 + \frac{\partial z}{\partial x} (3xz^2 - 2y) = 0$$

$$\frac{\partial z}{\partial x} (3xz^2 - 2y) = -y - z^3$$

$$\frac{\partial z}{\partial x} (2y - 3xz^2) = y + z^3$$

$$\frac{\partial z}{\partial x} = \frac{y + z^3}{2y - 3xz^2}$$

At $(1, 1, 1)$

$$\frac{\partial z}{\partial x} \Big|_{(1, 1, 1)}$$

$$= \frac{1 + 1}{2(1) - 3(1)(1)} = \frac{2}{2 - 3} = -2$$

Example: Express V_x in terms of u and V if the equations $x = V \ln u$ and $y = u \ln v$. Define u and V as function of the independent variables x and y , and if V_x exists.

(3)

Situation

$$x = V \ln u \quad \text{and} \quad y = u \ln v$$

$$\frac{\partial v}{\partial x} = ?$$

$$\frac{\partial}{\partial x} (x) = \frac{\partial}{\partial x} (V \ln u)$$

$$1 = V \frac{\partial}{\partial x} (\ln u) + \ln u \frac{\partial}{\partial x} (V)$$

$$= V \left(\frac{1}{u} \cdot \frac{\partial u}{\partial x} \right) + \ln u \frac{\partial V}{\partial x}$$

$$1 = \frac{V}{u} \frac{\partial u}{\partial x} + \ln u \frac{\partial V}{\partial x}$$

Now

$$\frac{\partial y}{\partial x} = \frac{\partial}{\partial x} (u \ln v) \quad \text{or} \quad \frac{V}{u} \frac{\partial u}{\partial x} = 1 - \ln u \frac{\partial V}{\partial x}$$

$$\frac{\partial y}{\partial x} = u \frac{\partial}{\partial x} \ln v + \ln v \frac{\partial}{\partial x} (u)$$

$$0 = u \cdot \frac{1}{v} \cdot \frac{\partial v}{\partial x} + \ln v \frac{\partial u}{\partial x}$$

$$= \frac{u}{v} \frac{\partial v}{\partial x} + \ln v \frac{\partial u}{\partial x}$$

$$\frac{u}{v} \frac{dv}{dx} = - \ln v \frac{\partial u}{\partial x}$$

$$\frac{\partial v}{\partial x} = -\frac{v}{u} \ln v \frac{\partial u}{\partial x}$$

$$= -\ln v \left(\frac{v}{u} \frac{\partial u}{\partial x} \right)$$

$$\frac{\partial v}{\partial x} = -\ln v \left(1 - \ln u \frac{\partial v}{\partial x} \right)$$

$$= -\ln v + \ln v \ln u \frac{\partial v}{\partial x}$$

$$\frac{\partial v}{\partial x} = -\ln v + \ln v \ln u \frac{\partial v}{\partial x}$$

$$\frac{\partial v}{\partial x} - \ln v \ln u \frac{\partial v}{\partial x} = -\ln v$$

$$\frac{\partial v}{\partial x} (1 - \ln v \ln u) = -\ln v$$

$$\frac{\partial v}{\partial x} = \frac{-\ln v}{1 - \ln v \ln u}$$

$$= \frac{1}{\frac{1}{\ln v} - \ln u}$$

$$=$$

Partial Derivatives of still higher order

(5)

$$\frac{\partial^3 f}{\partial x \partial y^2} = f_{yyx}$$

$$\frac{\partial^4 f}{\partial x^2 \partial y^2} = f_{yyxx}$$

and so on. As with second order derivatives, the order of differentiation is immaterial as long as all the derivatives through the order in question are continuous.

Example: Find f_{xyz} if

$$f(x, y, z) = 1 - 2xy^2z + x^2y$$

$$\frac{\partial^3 f}{\partial z \partial y \partial x} = f_{xyz}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (1 - 2xy^2z + x^2y)$$

$$= -4xy^2 + x^2$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (-4xy^2 + x^2)$$

$$= -4yz + 2x$$

Now

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x \partial y} \right) = \frac{\partial}{\partial y} (-4yz + 2x)$$

$$= -4z = \frac{\partial^2 f}{\partial y \partial x \partial y}$$

$$\frac{\partial^4 f}{\partial z \partial y \partial x \partial y} = \frac{\partial}{\partial z} (4z) = 4$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} (1 - 2xy^2z + x^2y)$$

$$= -2xy^2$$

$$\frac{\partial^2 f}{\partial x \partial z} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial z} \right) = \frac{\partial}{\partial x} (-2xy^2) = -2y^2$$

$$\frac{\partial^3 f}{\partial y \partial x \partial z} = \frac{\partial}{\partial y} \left(\frac{\partial^2 f}{\partial x \partial z} \right) = \frac{\partial}{\partial y} (-2y^2)$$

$$= -4y$$

$$\frac{\partial^4 f}{\partial y \partial y \partial x \partial z} = \frac{\partial}{\partial y} (-4y) = -4$$

order doesn't matter
Answers remain same.

Example: find the value of fifth-order partial derivative $\frac{\partial^5 f}{\partial x^2 \partial y^3}$ of $f(x, y) = y^2 + y(\sin x - x^4)$ (4)

$$\frac{\partial^5 f}{\partial x^2 \partial y^3} = \frac{\partial f}{\partial x^2 \partial y^3} = \frac{\cancel{\partial f}}{\cancel{\partial y}} = \frac{\partial}{\partial y} (y^2 + y \sin(x - x^4))$$

$$= 2y + \sin(x - x^4)$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (2y + \sin(x - x^4)) = 2$$

$$\frac{\partial^3 f}{\partial y^3} = \frac{\partial}{\partial y} \left(\frac{\partial^2 f}{\partial y^2} \right) = \frac{\partial}{\partial y} (2) = 0$$

$$\frac{\partial^4 f}{\partial x \partial y^3} = 0 \quad , \quad \frac{\partial^5 f}{\partial x^2 \partial y^3} = 0$$

Example: Find the value of fifth-order partial derivative $\frac{\partial^5 f}{\partial x^2 \partial y^3}$ of

$$f(x, y) = x e^{y^2/2}$$

$$\frac{\partial^5 f}{\partial x^2 \partial y^3} = \frac{\partial^2 f}{\partial y^3} = \frac{\partial}{\partial y} \left(x e^{y^2/2} \right)$$

$$= x \frac{\partial}{\partial y} (e^{y^2/2}) = x e^{y^2/2} \cdot \frac{\partial}{\partial y} (y^2/2)$$

$$= x e^{y^2/2} \cdot y$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (x e^{y^2/2} y) = x \left[e^{y^2/2} \frac{\partial}{\partial y} (y) + \right.$$

$$\left. y \cdot \frac{\partial}{\partial y} e^{y^2/2} \right]$$

$$= x y (e^{y^2/2} y) + x e^{y^2/2} = x y^2 e^{y^2/2} + x e^{y^2/2}$$

$$\frac{\partial^3 f}{\partial y^3} = \frac{\partial}{\partial y} \left(\frac{\partial^2 f}{\partial y^2} \right) = \frac{\partial}{\partial y} (x y^2 e^{y^2/2} + x e^{y^2/2})$$

$$x \left[2y \cdot e^{y^2/2} + e^{y^2/2} \cdot y \cdot y^2 + x e^{y^2/2} \cdot y \right]$$

$$\frac{\partial^3 f}{\partial y^3} = x \left[e^{y^2/2} y [2 + y^2 + 1] \right]$$

$$= x y e^{y^2/2} (3 + y^2)$$

$$\frac{\partial^4 f}{\partial x \partial y^3} = \frac{\partial}{\partial x} \left(2ye^{y^2/2} + y^2 e^{y^2/2} + \cancel{xye^{y^2/2}} \right) \textcircled{1}$$

$$= 2ye^{y^2/2} + y^2 e^{y^2/2} + \cancel{xye^{y^2/2}}$$

$$\frac{\partial^5 f}{\partial x^2 \partial y^3} = 0 + 0 + \cancel{xye^{y^2/2}}$$

$$\frac{\partial^5 f}{\partial x^2 \partial y^3} \text{ if } f(x,y) = xe^{y^2/2}$$

$$\Rightarrow \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (xe^{y^2/2}) = xe^{y^2/2} \cdot \frac{\partial y}{\partial y} = xy e^{y^2/2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} (xy e^{y^2/2}) = x \frac{\partial}{\partial y} (ye^{y^2/2})$$

$$= x \left[y \frac{\partial}{\partial y} (e^{y^2/2}) + e^{y^2/2} \cdot \frac{\partial y}{\partial y} \right]$$

$$= x \left(y e^{y^2/2} \cdot y + e^{y^2/2} \right)$$

$$= x \left(y^2 e^{y^2/2} + e^{y^2/2} \right) = xy^2 e^{y^2/2} + xe^{y^2/2}$$

$$\frac{\partial^3 f}{\partial y^3} = x \left(2y \cdot e^{y^2/2} + y \cdot y e^{y^2/2} + e^{y^2/2} \cdot y \right)$$

$$= x \left(2ye^{y^2/2} + y^2 e^{y^2/2} + ye^{y^2/2} \right)$$

$$\frac{\partial^4}{\partial x \partial y^3} = \frac{\partial}{\partial x} \left(\frac{\partial f^3}{\partial y^3} \right) = \frac{\partial}{\partial x} \left(x \left(2y e^{\frac{y^2}{2}} + y^3 e^{\frac{y^2}{2}} + y^2 e^{\frac{y^2}{2}} \right) \right)$$

$$= 2y e^{\frac{y^2}{2}} + y^3 e^{\frac{y^2}{2}} + e^{\frac{y^2}{2}} y$$

$$\frac{\partial^5}{\partial x^2 \partial y^3} = \frac{\partial}{\partial x} \left(\frac{\partial^4}{\partial x \partial y^3} \right) = 0 + 0 + 0 = 0$$

Rough work

$$\frac{d}{dz} \left(\frac{u}{z^3} \right) = \frac{1}{z^3} \frac{du}{dz} - \frac{3u}{z^4}$$

$$\frac{d}{dz} \left(\frac{u}{z^3} \right) = \frac{1}{z^3} \frac{du}{dz} - \frac{3u}{z^4}$$

$$\frac{d}{dz} \left(\frac{u}{z^3} \right) = \frac{1}{z^3} \frac{du}{dz} - \frac{3u}{z^4}$$

Example: Show that function

(11)

$$f(x,y) = e^{-2y} \cos 2x$$

satisfies Laplace equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

solution:-

$$\begin{aligned} \frac{\partial}{\partial x} (e^{-2y} \cos 2x) &= e^{-2y} (-\sin 2x \cdot 2) \\ &= -2e^{-2y} \sin 2x \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} (-2e^{-2y} \sin 2x) \\ &= -2e^{-2y} \cos 2x \cdot 2 \end{aligned}$$

$$\frac{\partial^2 f}{\partial x^2} = -4e^{-2y} \cos 2x$$

$$\begin{aligned} \text{Now } \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \\ &= \frac{\partial}{\partial y} (e^{-2y} \cos 2x) \\ &= e^{-2y} (-2) \cos 2x \\ &= -2e^{-2y} \cos 2x \end{aligned}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (-2e^{-2y} \cos 2x)$$

$$= -2e^{-2y}(-2) \cos 2x$$

$$\frac{\partial^2 f}{\partial y^2} = +4e^{-2y} \cos 2x$$

Now

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

$$\underline{-4e^{-2y} \cos 2x} + \underline{4e^{-2y} \cos 2x} = 0$$

$$0 = 0$$

Example: Show that $f(x, y, z) = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$ satisfies the Laplace equation.

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

$$\text{solution: } f(x, y, z) = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \right) \Rightarrow \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$= -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} (2x)$$

(13)

$$\frac{\partial}{\partial x} = -x (x^2 + y^2 + z^2)^{-\frac{3}{2}}$$

Again

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(-x (x^2 + y^2 + z^2)^{-\frac{3}{2}} \right)$$

$$= -x \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-\frac{3}{2}} + (x^2 + y^2 + z^2)^{-\frac{3}{2}} \frac{\partial}{\partial x} (-x)$$

$$= -x \cdot -\frac{3}{2} (x^2 + y^2 + z^2)^{-\frac{5}{2}} \cdot 2x + (x^2 + y^2 + z^2)^{-\frac{3}{2}} (-1)$$

$$= +3x^2 (x^2 + y^2 + z^2)^{-\frac{5}{2}} - (x^2 + y^2 + z^2)^{-\frac{3}{2}}$$

Now

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \right) \Rightarrow \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$= -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} (2y) = -y (x^2 + y^2 + z^2)^{-\frac{3}{2}} = \frac{\partial f}{\partial y}$$

Again

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} \left(-y (x^2 + y^2 + z^2)^{-\frac{3}{2}} \right)$$

$$= -y \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-\frac{3}{2}} + (x^2 + y^2 + z^2)^{-\frac{3}{2}} \frac{\partial}{\partial y} (-y)$$

$$= -y \left(-\frac{3}{2} (x^2 + y^2 + z^2)^{-\frac{5}{2}} (2y) \right) + (x^2 + y^2 + z^2)^{-\frac{3}{2}} (-1)$$

$$= +3y^2 (x^2 + y^2 + z^2)^{-\frac{5}{2}} - (x^2 + y^2 + z^2)^{-\frac{3}{2}}$$

similarly

$$\frac{\partial f}{\partial z^2} = \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} \right) =$$

$$= 3z^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

Now

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

$$\Rightarrow 3x^2 + 3y^2 + 3z^2 (x^2+y^2+z^2)^{-5/2} - 3(x^2+y^2+z^2)^{-3/2}$$

$$= 3(x^2+y^2+z^2)^{-5/2+1} - 3(x^2+y^2+z^2)^{-3/2}$$

$$= 3(x^2+y^2+z^2)^{-3/2} - 3(x^2+y^2+z^2)^{-3/2}$$

$$= \frac{3(x^2+y^2+z^2)^{-3/2} - 3(x^2+y^2+z^2)^{-3/2}}{}$$

$$= 0$$

Q.1 show that the function $w = \sin(x+ct) + \cos(2x+2ct)$ is solution of the wave equation (15)

Solution

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2}$$

$$\frac{\partial^2 w}{\partial t^2} - c^2 \frac{\partial^2 w}{\partial x^2} = 0$$

$$w(x,t) = \sin(x+ct) + \cos(2x+2ct)$$

$$\begin{aligned} \frac{\partial w}{\partial t} &= \frac{\partial}{\partial t} (\sin(x+ct)) + \frac{\partial}{\partial t} (\cos(2x+2ct)) \\ &= \cos(x+ct) \frac{\partial}{\partial t} (ct) + (-\sin(2x+2ct)) \frac{\partial}{\partial t} (2ct) \end{aligned}$$

$$= c \cos(x+ct) - 2c \sin(2x+2ct)$$

Again

$$\frac{\partial^2 w}{\partial t^2} = \frac{\partial}{\partial t} \left(\frac{\partial}{\partial t} (c \cos(x+ct) - 2c \sin(2x+2ct)) \right)$$

$$= \frac{\partial}{\partial t} \left(c \cos(x+ct) - 2c \frac{\partial}{\partial t} (\sin(2x+2ct)) \right)$$

$$= c (-\sin(x+ct)) \frac{\partial}{\partial t} (x+ct) - 2c \cos(2x+2ct) \frac{\partial}{\partial t} (2x+2ct)$$

$$= -c^2 \sin(x+ct) - 4c^2 \cos(2x+2ct)$$

$$\begin{aligned}\frac{\partial w}{\partial x} &= \frac{\partial}{\partial x} (\sin(x+ct)) + \frac{\partial}{\partial x} (\cos(2x+2ct)) \\ &= \cos(x+ct)(1) + (-\sin(2x+2ct)) \frac{\partial}{\partial x} (2x+2ct) \\ &= \cos(x+ct) - 2\sin(2x+2ct)\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 w}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial x} \right) \\ &= \frac{\partial}{\partial x} (\cos(x+ct) - 2\sin(2x+2ct)) \\ &= -\sin(x+ct) - 2\cos(2x+2ct)(2)\end{aligned}$$

$$\begin{aligned}&= -\sin(x+ct) - 4\cos(2x+2ct) \\ \text{Now } \frac{\partial^2 w}{\partial x^2} &= -(\sin(x+ct) - 4\cos(2x+2ct))\end{aligned}$$

$$\frac{\partial^2 w}{\partial t^2} - c^2 \frac{\partial^2 w}{\partial x^2} = 0$$

$$\frac{c^2 \sin(x+ct) - 4c^2 \cos(2x+2ct)}{c^2} - \frac{c^2 \sin(x+ct) + 4c^2 \cos(2x+2ct)}{c^2} = 0$$

$$0 = 0$$

2. show that the fo

$$u(x,t) = 5 \cos(2x + 2ct) + e^{x+ct}$$

is solution of the wave equation.

(12)